

RAILGUARD

==>Problem Statement:

Railway authorities need an intelligent system to **identify, predict, rank, and prioritize railway risks** so that the most critical issues are addressed first with limited maintenance resources.

1. Project Overview

- RailGuard is an AI-powered decision-support platform for railway maintenance.
- It does not replace existing inspection systems. It combines their outputs with traffic, weather, maintenance history and network information.
- Its main goal is to answer: Which railway risk should be handled first, why, what happens if maintenance is delayed, and how should limited resources be used?
- Core flow: Detect → Predict → Assess → Rank → Explain → Simulate → Optimize → Recommend → Learn.

2. Problem Statement

- Railway networks generate large amounts of inspection and operational data, but maintenance teams have limited time, budget and manpower.
- The same defect can create different levels of risk depending on traffic, weather, track condition, history and network importance.
- RailGuard addresses the gap between detecting a problem and deciding the most effective maintenance action.

3. Core Risk Engine

- Generates a Dynamic Risk Score from 0–100 using defect severity, failure probability, operational impact, traffic exposure, deterioration rate, weather, track age, maintenance history and network criticality.
- Classifies sections as Low, Medium, High or Critical.
- Ranks all detected and predicted risks across the network.
- Provides a clear priority such as Immediate, Urgent, Planned or Monitor.

4. Predictive Intelligence

- Predicts how risk will change over 24 hours, 3 days, 7 days, 14 days or longer.
- Calculates Risk Velocity to identify sections whose risk is increasing unusually fast.
- Estimates failure probability and potential impact separately, then combines them into an overall risk assessment.
- Generates early warnings for tracks predicted to become critical.

5. Explainable AI

- Shows why a track received its risk score and priority.
- Displays major contributors such as defect severity, traffic, deterioration, weather and previous failures.
- Answers: Why is this track risky? Why is it ranked above another track?

6. What-If & Digital-Twin-Inspired Simulation

- Allows engineers to compare maintenance today, maintenance after a delay and no intervention.

- Simulates conditions such as heavy rainfall, increased traffic or track closure.
- Shows how each scenario can change track and network risk.
- Helps estimate the consequence of delaying maintenance.

7. Maintenance & Budget Optimization

- Recommends actions such as inspection, preventive maintenance, detailed investigation, monitoring or scheduled intervention.
- Optimizes which maintenance tasks should be selected when budget or manpower is limited.
- Uses expected risk reduction, cost, urgency and network impact.
- Can calculate risk reduction per unit of maintenance cost.

8. Network Intelligence

- Represents railway sections, stations and junctions as a network.
- Considers route importance, train volume and availability of alternative routes.
- Identifies high-impact sections where a failure could affect connected routes.
- Future versions can simulate cascading failures across the network.

9. Recurring Failure & Root-Cause Intelligence

- Detects repeated failures at the same location.
- Flags sections where repeated repairs are not sufficiently reducing risk.
- Analyzes historical, environmental and operational patterns to identify possible contributing factors.
- Can recommend deeper investigation instead of repeated routine repairs.

10. Maintenance Feedback & Learning

- Compares predicted risk before maintenance with actual risk after maintenance.
- Measures maintenance effectiveness.
- Uses verified outcomes as feedback for future model improvement.
- Shows prediction confidence and can request additional inspection when confidence is low.

11. Human-in-the-Loop

- RailGuard provides recommendations rather than autonomous railway-control decisions.
- Authorized engineers review, approve, modify or reject recommendations.
- Human decisions and maintenance outcomes can be stored as feedback.

12. Advanced Features

- Emergency combined-risk detection for dangerous combinations such as severe defects, heavy traffic, rapid deterioration and extreme weather.
- Network Health Score for a quick overview of overall railway condition.
- Risk map showing Critical, High, Medium and Low sections.
- Natural-language assistant for questions such as: Which five sections need attention today?
- Automated daily risk brief containing network health, critical risks and recommended priorities.

13. Proposed Technology

- Frontend: React.js.
- Backend: Python with FastAPI or Flask.
- Data and ML: Pandas, NumPy, Scikit-learn and XGBoost.
- Optimization: OR-Tools or PuLP.
- Database: PostgreSQL or MySQL.
- Maps and visualization: OpenStreetMap with a suitable mapping and charting library.
- Deployment: Vercel, Render or equivalent cloud services.

14. SIH MVP

- Dynamic Risk Score.
- Future Risk Prediction.
- Risk Velocity.
- Network-wide Risk Ranking.
- Explainable AI.
- Risk Map and Dashboard.
- What-If Maintenance Simulator.
- Basic Budget Optimization.
- Recurring Failure Detection.
- Human-approved maintenance recommendation.

15. Future Scope

- Network risk graph and cascade-failure simulation.
- Digital-Twin-inspired railway simulation.
- Advanced budget and manpower optimization.
- Continuous model learning and maintenance-effectiveness analysis.
- Natural-language railway assistant.
- Integration with real outputs from existing railway inspection systems.

16. Key Differentiator

- Existing railway systems are specialized for inspection, measurement, monitoring or defect detection.
- RailGuard is positioned as a decision-intelligence layer that combines information from multiple sources.
- Existing systems help answer: What was detected and what is the condition?
- RailGuard aims to answer: What is the overall risk, which problem should be handled first, why, what happens if we wait, and how should limited resources be allocated?
- USP: From Track Data to Maintenance Decisions.

17. Final Concept

- RailGuard transforms railway data into predictive, explainable and resource-aware maintenance decisions.
- Its strongest value is not simply detecting railway defects, but prioritizing current and emerging risks and recommending the best response.
- Final workflow: Detect → Predict → Rank → Explain → Simulate → Optimize → Recommend → Learn.